RWANDA POLYTECHNIC

MACHINE LEARNING ASSIGNMENT

Rwanda House Prices Prediction

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2026

1. Introduction

This assignment applies machine learning to predict house prices in Rwanda. A model was trained on housing data and then deployed as a web application so that anyone can enter house details and receive a predicted price.

2. Project Links

GitHub repository (source code, dataset and notebook):

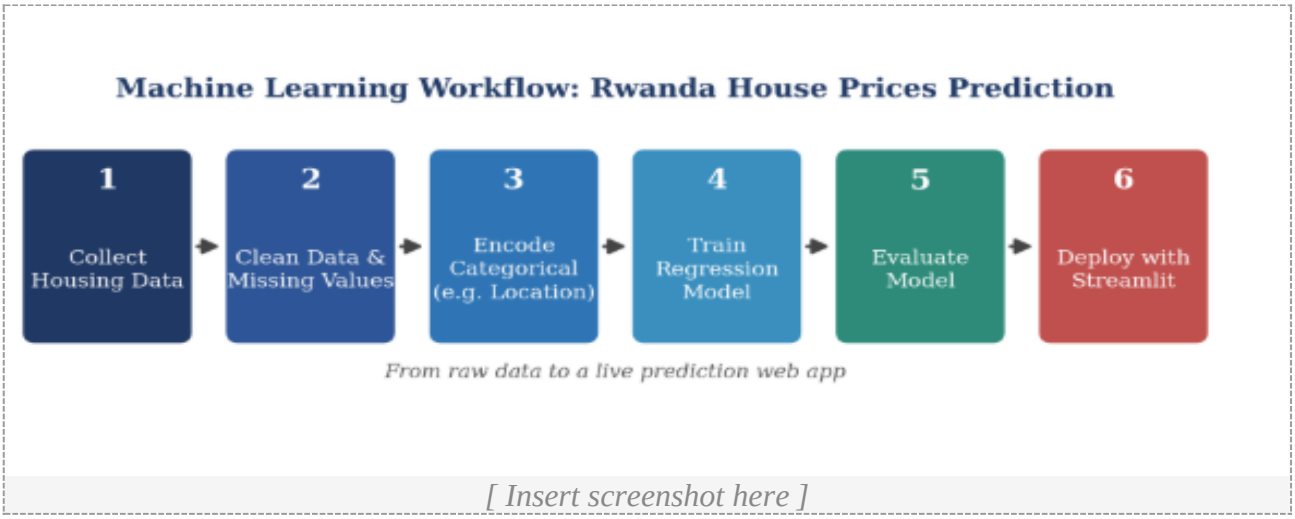
https://github.com/tuyishimiresaphina01-ship-it/RWANDA-HOUSE-PRICES-PREDICTION-

Live Streamlit application (prediction app):

https://zrpwzmcdna4bjan2bfyte6.streamlit.app/

3. Methodology

The work followed the usual machine learning steps: collecting the housing data, cleaning it and handling missing values, encoding categorical features such as location, training a regression model, evaluating it, and deploying it with Streamlit.



[Insert screenshot here]

Figure 1: GitHub repository

4. Data and Model

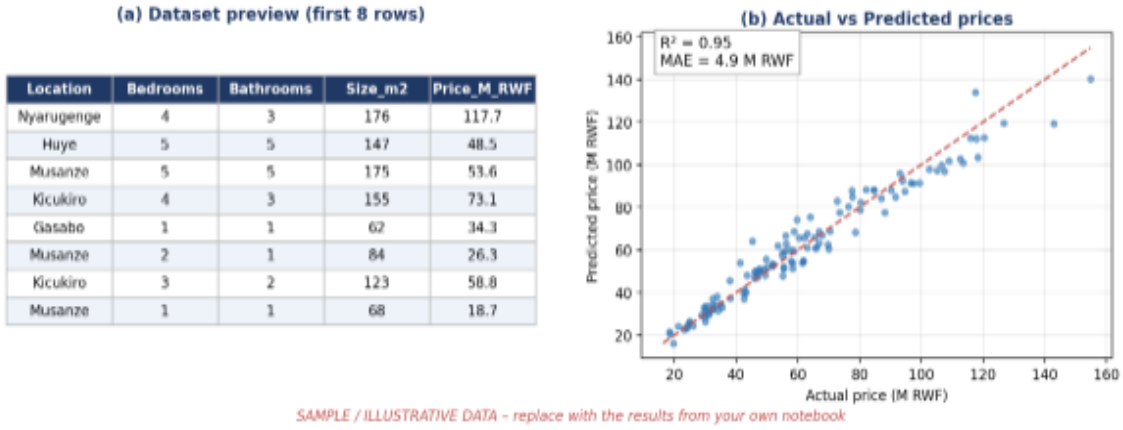
The dataset contains house features (for example location, size and number of rooms) together with the price. The data was split into training and testing sets, and the model was evaluated on the test set to check how close its predictions are to real prices.

[Insert screenshot here]

Figure 2: Dataset preview / model results

5. Web Application

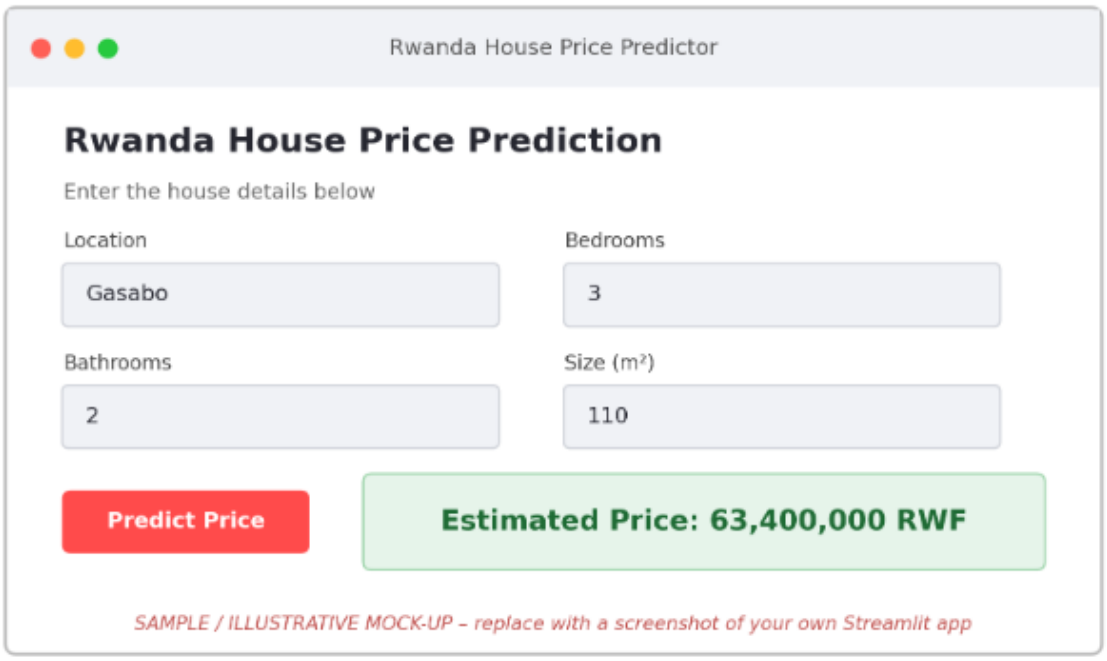
The Streamlit app lets a user enter the house details and instantly shows the predicted price.



Location	Bedrooms	Bathrooms	Size_m2	Price_M_RWF
Nyarugenge	4	3	176	117.7
Huye	5	5	147	48.5
Musanze	5	5	175	53.6
Kicukiro	4	3	155	73.1
Gasabo	1	1	62	34.3
Musanze	2	1	84	26.3
Kicukiro	3	2	123	58.8
Musanze	1	1	68	18.7

[Insert screenshot here]

Figure 3: Streamlit app interface



[Insert screenshot here]

Figure 4: Example prediction output

6. Conclusion

The project shows how machine learning can estimate house prices in Rwanda and how a trained model can be made easy to use through a web app. It could be improved with more data and further model tuning.